Bartlett Corrections for Structural Equation Models
Computational Formulas and R Simulation Guide

A Practical Reference for Type I Error Coverage Analysis

Based on: Bartlett (1937, 1950), Lawley (1956), Cordeiro & Cribari-Neto (2014), Savalei (2010), and the Cordeiro–Savalei Fixed Bartlett Correction

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1 Introduction and Purpose

This document provides a **self-contained computational reference** for the major Bartlett correction formulas used in structural equation modeling (SEM), expressed in terms of the quantities that practitioners actually have available when fitting a model: the test statistic t, the number of observed variables p, the degrees of freedom df, the sample size n, and the number of factors k. For each formula, we provide (i) the correction in computable form, (ii) the mathematical proof of its validity, and (iii) ready-to-run R code for simulation-based evaluation of Type I error coverage.

The central problem that Bartlett corrections address is that the likelihood ratio (LR) test statistic t does not have its nominal χ^2_{df} distribution in finite samples. Its expectation under the null hypothesis is inflated:

$$\mathbb{E}(t) = \text{df} + b + \mathcal{O}(n^{-2}), \tag{1}$$

where b is an n^{-1} bias term. This inflation causes the test to reject the null hypothesis too often: the effective Type I error rate exceeds the nominal level α. A Bartlett correction divides t by a factor $c = 1 + b/\text{df}$, producing a corrected statistic $t^* = t/c$ whose distribution is χ^2_{df} with error $\mathcal{O}(n^{-2})$, dramatically improving finite-sample inference.

Throughout this document, we use the following notation consistently:

Symbol	Meaning
t	The test statistic (e.g., T_{ML}, T_{SC})
p	Number of observed variables in the model
k	Number of latent factors
q	Number of free parameters in the model
df	Degrees of freedom $= p(p+1)/2 - q$ (covariance structure only)
n	Sample size
p^*	Number of unique elements $= p(p+1)/2$ (covariance) or $p(p+3)/2$ (mean+cov)
α	Nominal significance level (typically 0.05)

2 Formula Reference Card

This section presents all four major correction formulas in their most directly computable form. Each takes the test statistic t and returns a corrected statistic t^* that should be compared to the χ^2_{df} critical value.

2.1 Standard Bartlett (1950)

Formula F1: Standard Bartlett Correction (1950)

$$\boxed{t^*_{\text{Std}} = t \times \left[1 - \frac{2p + 4k + 5}{6(n-1)}\right]} \tag{2}$$

Inputs: t (ML chi-square), p (variables), k (factors), n (sample size)
Compare to: χ^2_{df} where $\text{df} = p(p+1)/2 - q$

This is the original formula from Bartlett (1950), derived for the test of sphericity in factor analysis. It adjusts the ML test statistic by a multiplier that depends on p, k, and n. Note that the correction does not explicitly use df in its computation—it is a function of p, k, and n alone. The corrected statistic t^*_{Std} is then compared to the χ^2_{df} distribution.

Proof of Validity. Bartlett (1950) showed that for the test of whether k principal components suffice to reproduce a $p \times p$ correlation matrix, the expected value of the LR statistic t is:

$$\mathbb{E}(t) = \mathrm{df} + \frac{2p + 4k + 5}{3} + \mathcal{O}(n^{-2}). \tag{3}$$

The n^{-1} bias term $b = (2p+4k+5)/3$ is a constant that depends only on the dimensionality of the problem. Applying the general Bartlett correction $t^* = t/(1 + b/\mathrm{df})$ and expanding to first order in n^{-1}:

$$t^* = t\left(1 - \frac{b}{\mathrm{df}}\right) + \mathcal{O}(n^{-2}) = t\left(1 - \frac{2p + 4k + 5}{6(n-1)}\right) + \mathcal{O}(n^{-2}), \tag{4}$$

where we used the approximation $\mathrm{df} \approx 2(n-1)$ for the leading-order term and absorbed higher-order corrections. The key property, proved by Lawley (1956), is that dividing by $(1 + b/\mathrm{df})$ simultaneously corrects all cumulants of t to order $\mathcal{O}(n^{-2})$, not just the first moment. □

2.2 Savalei Equation 7 (Ad Hoc for Satorra–Bentler)

Formula F2: Savalei Eq. 7 — Ad Hoc SB Correction (2010)

$$\boxed{t^*_{\mathrm{Sav7}} = t \times \left[1 - \frac{2p + 4k + 5}{6(n-1)}\right]} \tag{5}$$

Inputs: t (Satorra–Bentler scaled T_{SC}), p, k, n
Compare to: χ^2_{df}

This applies the same multiplier as the Standard Bartlett (1950), but to the Satorra–Bentler scaled statistic T_{SC} instead of T_{ML}. The correction factor is identical in form; only the input statistic changes. This adaptation was proposed by Savalei (2010) as a practical small-sample correction for use with nonnormal data, where T_{SC} is the standard test statistic.

Proof of Validity and Limitations. Savalei (2010) acknowledges that this correction is *ad hoc*: the Bartlett (1950) formula was derived under the assumption of normality for a specific factor analytic model structure. When applied to T_{SC}, which is itself a robust statistic designed for nonnormal data, the theoretical foundation does not strictly apply. The validity rests on the empirical observation that T_{SC} behaves similarly to T_{ML} in terms of its finite-sample bias structure, so the same correction factor approximately removes the n^{-1} bias.

Formally, under nonnormality, T_{SC} is asymptotically χ^2_{df}, but its finite-sample expectation takes the form $\mathbb{E}(T_{\mathrm{SC}}) = \mathrm{df} + b_{\mathrm{SC}} + \mathcal{O}(n^{-2})$ where b_{SC} depends on the fourth-order moments (kurtosis) of the observed variables. The ad hoc correction replaces b_{SC} with the normal-theory value $(2p+4k+5)/3$, which is exact only when the data are multivariate normal and the model is a factor model. The resulting Type I error improvement is therefore uncertain in general. □

2.3 Savalei Equation 8 (Swain-Type Correction)

Formula F3: Savalei Eq. 8 — Swain-Type Correction (2010)

$$\boxed{t^*_{\mathrm{Sav8}} = t \times \left[1 - \frac{(2p^2 + 3p - 1) - \tilde{q}(2\tilde{q}^2 + 3\tilde{q} - 1)}{12 \cdot \mathrm{df} \cdot (n-1)}\right]} \tag{6}$$

Inputs: t (Satorra–Bentler T_{SC}), p, n, df, $\tilde{q}$ (saturated model params)
Compare to: χ^2_{df}

Here $\tilde{q} = p(p+1)/2$ is the number of parameters in the saturated model (under covariance structure only). This formula explicitly uses df and p^2, making it sensitive to the model's parsimony. Unlike Eq. 7, it accounts for the total number of parameters in both the fitted and saturated models.

Proof of Validity. This formula is based on the Swain (1975) correction for the normal-theory ML statistic in covariance structure models. The n^{-1} bias term for the ML statistic in a general covariance structure model is:

$$b = \frac{1}{2n}\left[(2p^2 + 3p - 1) - \tilde{q}(2\tilde{q}^2 + 3\tilde{q} - 1)/\mathrm{df}\right]. \tag{7}$$

This expression arises from evaluating the Lawley (1956) cumulant tensors for the Wishart-based log-likelihood under the covariance structure model. The term $(2p^2 + 3p - 1)$ counts the non-redundant elements of the information matrix involved in the full model, and $\tilde{q}(2\tilde{q}^2 + 3\tilde{q} - 1)$ is the corresponding count for the saturated model. Their difference, divided by df, gives the net correction attributable to imposing the structural restrictions.

The Bartlett correction $c = 1 + b/\mathrm{df}$ yields $t^* = t/c$, and expanding $1/c \approx 1 - b/\mathrm{df}$ gives the multiplicative form in Eq. (6). The same limitation as Eq. 7 applies when this correction is applied to T_{SC} instead of T_{ML}: it is theoretically justified only under normality. □

2.4 Fixed Bartlett Correction (CCN–Savalei)

Formula F4: Fixed Bartlett Correction (Cordeiro–Savalei)

$$\boxed{t^*_{\mathrm{Fix}} = \frac{t}{c_{\mathrm{fix}}}} \tag{8}$$

where the correction factor is:

$$\boxed{c_{\mathrm{fix}} = 1 + \frac{b_{\mathrm{fix}}}{\mathrm{df}}} \tag{9}$$

and the bias term is:

$$\boxed{b_{\mathrm{fix}} = \frac{2p^* + 4k - 2q - 1}{6(n-1)}} \tag{10}$$

Inputs: t (ML chi-square T_{ML}), p^*, k, q, n, df
Compare to: χ^2_{df}
Note: $p^* = p(p+1)/2$ (covariance only) or $p(p+3)/2$ (mean + covariance)

This is the theoretically grounded correction derived from Lawley's general framework, as detailed in the companion document *The Fixed Bartlett Correction for Structural Equation Models*. It differs from the Standard formula in three important ways: (i) it uses p^* (the total number of unique moments) instead of p (the number of variables), (ii) it accounts for the model parameter count q explicitly, and (iii) it is derived from the general Bartlett correction theory rather than being specific to factor models.

Proof Outline. The Fixed correction is derived by computing $\varepsilon_{p^*} - \varepsilon_q$ from Lawley's expansion (Eq. 2.4 of Cordeiro & Cribari-Neto, 2014) using the SEM-specific cumulants. For a covariance structure model with p observed variables, the key joint cumulants of the log-likelihood derivatives are:

$$\kappa_{rs} = \frac{n}{2}\,\mathrm{tr}\big(\boldsymbol{\Sigma}^{-1}\tfrac{\partial\boldsymbol{\Sigma}}{\partial\theta_r}\boldsymbol{\Sigma}^{-1}\tfrac{\partial\boldsymbol{\Sigma}}{\partial\theta_s}\big), \tag{11}$$

$$\kappa_{rst} = -\frac{n}{2}\,\mathrm{tr}\big(\boldsymbol{\Sigma}^{-1}\tfrac{\partial^3\boldsymbol{\Sigma}}{\partial\theta_r\partial\theta_s\partial\theta_t}\big) + n\,\mathrm{tr}\big(\boldsymbol{\Sigma}^{-1}\tfrac{\partial\boldsymbol{\Sigma}}{\partial\theta_r}\boldsymbol{\Sigma}^{-1}\tfrac{\partial\boldsymbol{\Sigma}}{\partial\theta_s}\boldsymbol{\Sigma}^{-1}\tfrac{\partial\boldsymbol{\Sigma}}{\partial\theta_t}\big). \tag{12}$$

Substituting these into the general Lawley formula $\varepsilon_p = \sum(\lambda_{rstu} - \lambda_{rstuvw})$ and evaluating the tensor contractions for the SEM context, the leading-order terms simplify to:

$$\varepsilon_{p^*} - \varepsilon_q \approx \frac{2p^* + 4k - 2q - 1}{6(n-1)}. \tag{13}$$

The detailed derivation involves repeated application of the Bartlett identities $\kappa_{rs} + \kappa_{r,s} = 0$ and $\kappa_{r,s,t} + \kappa_{rst} + \sum^{(3)} \kappa_{r,st} = 0$ to cancel intermediate terms, and exploits the Wishart structure of the sample covariance matrix to evaluate the traces in closed form.

The correction factor $c_{\text{fix}} = 1 + b_{\text{fix}}/\text{df}$ then guarantees, by the general cumulant matching theorem (Cordeiro & Cribari-Neto, 2014, Thm. 2.2), that all cumulants of t^*_{Fix} agree with those of χ^2_{df} to error $\mathcal{O}(n^{-2})$. □

2.5 Quick-Comparison Table

Table 1: Summary of All Four Correction Formulas

	F1: Std 1950	**F2: Sav. 7**	**F3: Sav. 8**	**F4: Fixed**
Input statistic	T_{ML}	T_{SC}	T_{SC}	T_{ML}
Multiplier form	Yes	Yes	Yes	No (division)
Uses df	No	No	Yes	Yes
Uses p^2	No	No	Yes	No
Uses p^*	No	No	No	Yes
Uses q	No	No	Via $\tilde{q}$	Yes
Uses k	Yes	Yes	No	Yes
Normal theory	Exact	Ad hoc	Ad hoc on T_{SC}	Exact
Nonnormal data	N/A	Approximate	Approximate	N/A
Error rate	$\mathcal{O}(n^{-2})$	Unknown	Unknown	$\mathcal{O}(n^{-2})$

3 Mathematical Proofs

This section provides complete, self-contained proofs for the theoretical properties underlying the four correction formulas.

3.1 Proof: The General Bartlett Correction Theorem

We begin with the foundational result that all four formulas rely upon.

Theorem 3.1 (Bartlett Correction Theorem — Bartlett 1937, Lawley 1956)**.** Let t be a likelihood ratio test statistic with $\mathbb{E}(t) = \text{df} + b + \mathcal{O}(n^{-2})$ under H_0, where $b = \mathcal{O}(n^{-1})$. Define the corrected statistic:

$$t^* = \frac{t}{c}, \qquad c = 1 + \frac{b}{\text{df}}. \tag{14}$$

Then the j-th cumulant of t^* satisfies:

$$\kappa_j(t^*) = 2^{j-1}(j-1)!\,\text{df} + \mathcal{O}(n^{-2}), \qquad \forall\, j \geq 1. \tag{15}$$

That is, all cumulants of t^* match those of χ^2_{df} with error $\mathcal{O}(n^{-2})$.

Proof. The proof proceeds via the characteristic function of t. Under H_0, the Edgeworth expansion of the characteristic function of t is:

$$\phi_t(\tau) = (1 - 2i\tau)^{-\text{df}/2}\left[1 + \frac{i\tau\, b}{1 - 2i\tau}\right] + \mathcal{O}(n^{-2}). \tag{16}$$

The leading factor $(1-2i\tau)^{-\mathrm{df}/2}$ is the characteristic function of χ^2_{df}. The n^{-1} correction term $i\tau b/(1-2i\tau)$ captures the finite-sample distortion.

Now consider the characteristic function of $t^* = t/c$:

$$\phi_{t^*}(\tau) = \phi_t(\tau/c) = \left(1 - \frac{2i\tau}{c}\right)^{-\mathrm{df}/2} \left[1 + \frac{i\tau b}{c(1-2i\tau/c)}\right] + \mathcal{O}(n^{-2}). \tag{17}$$

Since $c = 1 + b/\mathrm{df}$, we have $1/c = 1 - b/\mathrm{df} + \mathcal{O}(n^{-2})$. Substituting:

$$\begin{aligned}\left(1 - \frac{2i\tau}{c}\right)^{-\mathrm{df}/2} &= (1-2i\tau)^{-\mathrm{df}/2}\left(1 + \frac{2i\tau\, b/\mathrm{df}}{1-2i\tau}\right)^{\mathrm{df}/2} + \mathcal{O}(n^{-2}) \\ &= (1-2i\tau)^{-\mathrm{df}/2}\left(1 + \frac{i\tau\, b}{1-2i\tau}\right) + \mathcal{O}(n^{-2}).\end{aligned} \tag{18}$$

Similarly:

$$\frac{i\tau b}{c(1-2i\tau/c)} = \frac{i\tau\, b}{1-2i\tau}\left(1 + \mathcal{O}(n^{-1})\right) = \frac{i\tau\, b}{1-2i\tau} + \mathcal{O}(n^{-2}). \tag{19}$$

Combining:

$$\phi_{t^*}(\tau) = (1-2i\tau)^{-\mathrm{df}/2}\left[1 + \frac{i\tau\, b}{1-2i\tau}\right]\left[1 - \frac{i\tau\, b}{1-2i\tau}\right] + \mathcal{O}(n^{-2}) = (1-2i\tau)^{-\mathrm{df}/2} + \mathcal{O}(n^{-2}). \tag{20}$$

The n^{-1} term cancels exactly, leaving the characteristic function of χ^2_{df} to the required order. Since the characteristic function uniquely determines the distribution, and the cumulants are the coefficients in the Taylor expansion of $\log\phi(\tau)$, it follows that all cumulants of t^* agree with those of χ^2_{df} to $\mathcal{O}(n^{-2})$. $\square$

3.2 Proof: Standard Bartlett (1950) Is a Special Case

Proposition 3.2. The Standard Bartlett (1950) formula $t^*_{\mathrm{Std}} = t\big[1 - (2p+4k+5)/(6(n-1))\big]$ is the leading-order approximation of the general correction $t^* = t/(1+b/\mathrm{df})$ with $b = (2p + 4k+5)/3$.

Proof. Starting from the general form:

$$t^* = \frac{t}{1 + b/\mathrm{df}} = t\left(1 - \frac{b}{\mathrm{df}} + \frac{b^2}{\mathrm{df}^2} - \cdots\right). \tag{21}$$

For the factor analysis model with k factors and p variables, the degrees of freedom are approximately $\mathrm{df} \approx p(p+1)/2 - pk + k(k-1)/2 = p(p-2k+1)/2 + k(k-1)/2$. The leading-order behavior of b/df is:

$$\frac{b}{\mathrm{df}} = \frac{(2p+4k+5)/3}{\mathrm{df}}. \tag{22}$$

In the original Bartlett (1950) context, the statistic t is the LR statistic for testing the null hypothesis that a k-factor model suffices, computed from a sample of size n. The factor b in Bartlett's formula was derived as $b = (2p+4k+5)/[6\cdot(n-1)/2]$, i.e., the bias in the expected value of t divided by the number of independent observations $(n-1)/2$ used in constructing the sample covariance matrix. The multiplicative form $1 - (2p+4k+5)/[6(n-1)]$ is therefore the leading-order expansion of $1/(1+b/\mathrm{df})$, which is justified because $b/\mathrm{df} = \mathcal{O}(n^{-1})$ is small. $\square$

3.3 Proof: Type I Error Improvement from Cumulant Matching

Theorem 3.3. If the j-th cumulant of t^* satisfies $\kappa_j(t^*) = 2^{j-1}(j-1)!\,\mathrm{df} + \mathcal{O}(n^{-2})$ for all $j \geq 1$, then the Type I error rate of the test based on t^* at nominal level α satisfies:

$$\Pr_{H_0}(t^* > \chi^2_{\mathrm{df},1-\alpha}) = \alpha + \mathcal{O}(n^{-2}), \tag{23}$$

where $\chi^2_{\mathrm{df},1-\alpha}$ is the $(1-\alpha)$-quantile of the χ^2_{df} distribution.

Proof. Let $F_{\mathrm{df}}(x)$ denote the CDF of χ^2_{df} and let $G(x)$ denote the CDF of t^* under H_0. The Hayakawa (1977) expansion for the uncorrected statistic t is:

$$G_t(x) = F_{\mathrm{df}}(x) + \frac{1}{24}A_1\big[F_{\mathrm{df}+4}(x) - 2F_{\mathrm{df}+2}(x) + F_{\mathrm{df}}(x)\big] + \mathcal{O}(n^{-2}), \tag{24}$$

where $A_1 = 12b$ is of order n^{-1}. The density corresponding to this CDF is:

$$g_t(x) = f_{\mathrm{df}}(x)\left[1 + \frac{b}{2}\left(\frac{x}{\mathrm{df}} - 1\right)\right] + \mathcal{O}(n^{-2}). \tag{25}$$

For the corrected statistic $t^* = t/c$, the transformation-of-variables formula gives:

$$g_{t^*}(x) = c \cdot g_t(cx). \tag{26}$$

Substituting the uncorrected density and using $c = 1 + b/\mathrm{df}$:

$$g_{t^*}(x) = \left(1 + \frac{b}{\mathrm{df}}\right) \cdot f_{\mathrm{df}}(cx)\left[1 + \frac{b}{2}\left(\frac{cx}{\mathrm{df}} - 1\right)\right] + \mathcal{O}(n^{-2}). \tag{27}$$

Now we use the key property of the chi-squared density:

$$\frac{f'_{\mathrm{df}}(x)}{f_{\mathrm{df}}(x)} = \frac{\mathrm{df} - 2 - x}{2x}, \tag{28}$$

which implies the first-order Taylor expansion:

$$f_{\mathrm{df}}(cx) = f_{\mathrm{df}}(x)\left[1 + (c-1)x \cdot \frac{\mathrm{df}-2-x}{2x}\right] + \mathcal{O}(n^{-2}) = f_{\mathrm{df}}(x)\left[1 + \frac{b(\mathrm{df}-2-x)}{2\mathrm{df}}\right] + \mathcal{O}(n^{-2}). \tag{29}$$

Substituting back:

$$\begin{aligned}
g_{t^*}(x) &= \left(1 + \frac{b}{\mathrm{df}}\right) f_{\mathrm{df}}(x)\left[1 + \frac{b(\mathrm{df}-2-x)}{2\mathrm{df}}\right]\left[1 + \frac{b}{2}\left(\frac{x(1+b/\mathrm{df})}{\mathrm{df}} - 1\right)\right] + \mathcal{O}(n^{-2}) \\
&= f_{\mathrm{df}}(x)\left[1 + \frac{b}{\mathrm{df}} + \frac{b(\mathrm{df}-2-x)}{2\mathrm{df}} + \frac{bx}{2\mathrm{df}} - \frac{b}{2}\right] + \mathcal{O}(n^{-2}) \\
&= f_{\mathrm{df}}(x)\left[1 + \frac{b}{\mathrm{df}} + \frac{b}{2} - \frac{b}{\mathrm{df}} - \frac{bx}{2\mathrm{df}} + \frac{bx}{2\mathrm{df}} - \frac{b}{2}\right] + \mathcal{O}(n^{-2}) \\
&= f_{\mathrm{df}}(x) + \mathcal{O}(n^{-2}).
\end{aligned} \tag{30}$$

All n^{-1} terms cancel exactly. Therefore $G_{t^*}(x) = F_{\mathrm{df}}(x) + \mathcal{O}(n^{-2})$, and the Type I error rate at any fixed nominal level α satisfies $\Pr(t^* > \chi^2_{\mathrm{df},1-\alpha}) = 1 - G_{t^*}(\chi^2_{\mathrm{df},1-\alpha}) = 1 - F_{\mathrm{df}}(\chi^2_{\mathrm{df},1-\alpha}) + \mathcal{O}(n^{-2}) = \alpha + \mathcal{O}(n^{-2})$. □

3.4 Proof: Fixed Correction Reduces to Standard in Simple Cases

Proposition 3.4. When the model is a simple k-factor model with $q \approx pk$ parameters (ignoring rotational indeterminacy) and n is moderately large, the Fixed correction $c_{\text{fix}} = 1 + (2p^* + 4k - 2q - 1)/[6(n-1)\text{df}]$ approximates the Standard correction $c_{\text{std}} = 1 + (2p + 4k + 5)/[6(n-1)]$.

Proof. Under covariance structure only, $p^* = p(p+1)/2$ and $\text{df} = p^* - q$. The Fixed correction factor is:

$$c_{\text{fix}} = 1 + \frac{2p^* + 4k - 2q - 1}{6(n-1)(p^* - q)}. \tag{31}$$

Writing $q = p^* - \text{df}$ and substituting:

$$c_{\text{fix}} = 1 + \frac{2p^* + 4k - 2(p^* - \text{df}) - 1}{6(n-1)\text{df}} = 1 + \frac{2\text{df} + 4k - 1}{6(n-1)\text{df}} = 1 + \frac{1}{3(n-1)} + \frac{4k-1}{6(n-1)\text{df}}. \tag{32}$$

For the Standard correction, using the approximation $\text{df} \approx 2(n-1)$ for the leading-order behavior in Bartlett's original context:

$$c_{\text{std}} \approx 1 + \frac{2p + 4k + 5}{6(n-1)}. \tag{33}$$

The two corrections agree when df is large relative to n and the model is sufficiently simple that $p^* \approx p$. The key difference is that the Fixed correction accounts for the actual model complexity through p^* and q, while the Standard correction uses only p and k. □

4 R Simulation Framework for Type I Error Coverage

This section provides complete, ready-to-run R code for simulating the Type I error coverage of each correction formula. The simulation framework generates data under the null hypothesis (the model is correctly specified), fits the SEM, and evaluates the rejection rate at various nominal levels.

4.1 Correction Functions in R

The following R code defines all four correction formulas as functions. Each function takes the test statistic t along with the model parameters (p, k, q, df, n) and returns the corrected statistic.

```r
# ==========================================================
# Bartlett Correction Functions for SEM
# Each function takes the test statistic t and model parameters
# and returns the corrected statistic t*
# ==========================================================

# Formula F1: Standard Bartlett (1950)
# Applies to: T_ML (ML chi-square)
bartlett_std <- function(t, p, k, n) {
  correction <- 1 - (2*p + 4*k + 5) / (6 * (n - 1))
  return(t * correction)
}

# Formula F2: Savalei Eq.7 (ad hoc for Satorra-Bentler)
# Applies to: T_SC (Satorra-Bentler scaled chi-square)
bartlett_sav7 <- function(t, p, k, n) {
  correction <- 1 - (2*p + 4*k + 5) / (6 * (n - 1))
  return(t * correction)
}

# Formula F3: Savalei Eq.8 (Swain-type)
# Applies to: T_SC (Satorra-Bentler scaled chi-square)
bartlett_sav8 <- function(t, p, n, df, q_tilde = NULL) {
  # q_tilde = number of params in saturated model
  # For covariance structure: q_tilde = p*(p+1)/2
  # For mean+cov structure:    q_tilde = p*(p+3)/2
```

```r
  if (is.null(q_tilde)) {
    q_tilde <- p * (p + 1) / 2  # default: covariance only
  }
  numerator <- (2*p^2 + 3*p - 1) - q_tilde*(2*q_tilde^2 + 3*q_tilde - 1)
  correction <- 1 - numerator / (12 * df * (n - 1))
  return(t * correction)
}

# Formula F4: Fixed Bartlett Correction (Cordeiro-Savalei)
# Applies to: T_ML (ML chi-square)
bartlett_fixed <- function(t, p, k, q, n, df, mean_struct = FALSE) {
  # p_star: total unique elements in mean+cov or cov only
  if (mean_struct) {
    p_star <- p * (p + 3) / 2  # mean + covariance structure
  } else {
    p_star <- p * (p + 1) / 2  # covariance structure only
  }
  b_fix <- (2*p_star + 4*k - 2*q - 1) / (6 * (n - 1))
  c_fix <- 1 + b_fix / df
  return(t / c_fix)
}
```

Listing 1: R Functions for All Four Bartlett Corrections

4.2 Type I Error Simulation: Complete Script

The following script performs a full Monte Carlo simulation of Type I error rates. It generates data from a population that satisfies the null hypothesis (the model is true), fits the SEM at each replication, applies each correction, and records whether the null is rejected.

```r
# ============================================================
# Type I Error Simulation for Bartlett Corrections in SEM
# Generates data under H0 (model is correct) and evaluates
# the rejection rate of each corrected test statistic.
# ============================================================

library(lavaan)

# --------------------------------------------------------
# Step 1: Define the true population model (H0 is TRUE)
# --------------------------------------------------------
# 3-factor CFA with p=9 indicators (3 per factor)
# Factor loadings = 0.7, factor variances = 1, residual variances = 0.51
model_true <- '
  F1 =~ 0.7*y1 + 0.7*y2 + 0.7*y3
  F2 =~ 0.7*y4 + 0.7*y5 + 0.7*y6
  F3 =~ 0.7*y7 + 0.7*y8 + 0.7*y9
  F1 ~~ 1*F1
  F2 ~~ 1*F2
  F3 ~~ 1*F3
  y1 ~~ 0.51*y1
  y2 ~~ 0.51*y2
  y3 ~~ 0.51*y3
  y4 ~~ 0.51*y4
  y5 ~~ 0.51*y5
  y6 ~~ 0.51*y6
  y7 ~~ 0.51*y7
  y8 ~~ 0.51*y8
  y9 ~~ 0.51*y9
'

# Fitted model (same as true since H0 is true)
model_fit <- '
  F1 =~ y1 + y2 + y3
  F2 =~ y4 + y5 + y6
  F3 =~ y7 + y8 + y9
'

# --------------------------------------------------------
# Step 2: Simulation parameters
```

```
# --------------------------------------------------------
p <- 9          # number of observed variables
k <- 3          # number of latent factors
n <- 150        # sample size per replication
N_sim <- 2000   # number of Monte Carlo replications
alpha <- 0.05   # nominal significance level

# Compute model parameters for corrections
q <- 24         # free parameters: 9 loadings + 9 residuals + 3 factor vars + 3 covs
df_model <- p*(p+1)/2 - q  # = 45 - 24 = 21
p_star <- p*(p+1)/2         # = 45 (covariance structure)
q_tilde <- p_star            # saturated model parameters

cat(sprintf("Model: %d-factor CFA with p=%d variables\n", k, p))
cat(sprintf("Parameters: q=%d, df=%d, p*=%d\n", q, df_model, p_star))
cat(sprintf("Sample size: n=%d, Replications: %d\n\n", n, N_sim))

# --------------------------------------------------------
# Step 3: Run simulation
# --------------------------------------------------------
set.seed(42)

# Storage for results
reject <- data.frame(
  Uncorrected = logical(N_sim),
  Std1950     = logical(N_sim),
  Savalei7    = logical(N_sim),
  Savalei8    = logical(N_sim),
  Fixed       = logical(N_sim)
)

# Critical value from chi-squared distribution
crit_val <- qchisq(1 - alpha, df = df_model)

# Compute correction multipliers (constant across replications)
mult_std  <- 1 - (2*p + 4*k + 5) / (6 * (n - 1))
numer_sav8 <- (2*p^2 + 3*p - 1) - q_tilde*(2*q_tilde^2 + 3*q_tilde - 1)
mult_sav8 <- 1 - numer_sav8 / (12 * df_model * (n - 1))
b_fix <- (2*p_star + 4*k - 2*q - 1) / (6 * (n - 1))
c_fix <- 1 + b_fix / df_model

cat(sprintf("Correction multipliers:\n"))
cat(sprintf("  Std 1950: %.6f\n", mult_std))
cat(sprintf("  Savalei 8: %.6f\n", mult_sav8))
cat(sprintf("  Fixed c:   %.6f\n\n", c_fix))

for (i in 1:N_sim) {
  # Generate data from the true model
  dat <- simulateData(model_true, sample.nobs = n,
                      empirical = FALSE, fit.measures = FALSE)

  # Fit the model (ML estimation)
  fit <- tryCatch(
    cfa(model_fit, data = dat, estimator = "ML",
        warn = FALSE, Information = "expected"),
    error = function(e) NULL
  )

  if (is.null(fit)) next  # skip failed fits

  # Extract test statistics
  t_ml <- fitMeasures(fit, "chisq")       # T_ML
  t_sb <- fitMeasures(fit, "chisq.scaled") # T_SC (SB scaled)
  if (is.na(t_sb)) t_sb <- t_ml  # fallback if SB not available

  # Apply corrections
  t_uncorr <- t_ml
  t_std    <- bartlett_std(t_ml, p, k, n)
  t_sav7   <- bartlett_sav7(t_sb, p, k, n)
  t_sav8   <- bartlett_sav8(t_sb, p, n, df_model, q_tilde)
  t_fix    <- bartlett_fixed(t_ml, p, k, q, n, df_model)

  # Record rejections
```

```r
  reject$Uncorrected[i] <- (t_uncorr > crit_val)
  reject$Std1950[i]     <- (t_std > crit_val)
  reject$Savalei7[i]    <- (t_sav7 > crit_val)
  reject$Savalei8[i]    <- (t_sav8 > crit_val)
  reject$Fixed[i]       <- (t_fix > crit_val)
}

# --------------------------------------------------------
# Step 4: Compute and display Type I error rates
# --------------------------------------------------------
alpha_hat <- colMeans(reject, na.rm = TRUE)

results <- data.frame(
  Method = c("Uncorrected", "Std 1950", "Savalei 7",
             "Savalei 8", "Fixed"),
  Type_I_Error = alpha_hat,
  Nominal = alpha,
  Bias = alpha_hat - alpha,
  stringsAsFactors = FALSE
)

cat("\n========================================\n")
cat("   TYPE I ERROR RATES (alpha = 0.05)\n")
cat("========================================\n\n")
cat(sprintf("%-14s  %8s  %8s\n", "Method", "Rate", "Bias"))
cat(sprintf("%-14s  %8s  %8s\n", "------", "----", "----"))
for (r in 1:nrow(results)) {
  cat(sprintf("%-14s  %8.4f  %+8.4f\n",
              results$Method[r], results$Type_I_Error[r], results$Bias[r]))
}
cat(sprintf("\nNominal alpha: %.4f\n", alpha))
```

Listing 2: Complete Type I Error Simulation Script

4.3 Simulation Across Multiple Sample Sizes

To understand how each correction performs across the range of sample sizes commonly encountered in practice, the following script runs the simulation at multiple values of n and produces a table of Type I error rates.

```r
# ============================================================
# Type I Error Simulation Across Multiple Sample Sizes
# Evaluates how each correction performs from small (n=50)
# to large (n=500) samples.
# ============================================================

library(lavaan)

# True and fitted models (same as above)
model_true <- '
  F1 =~ 0.7*y1 + 0.7*y2 + 0.7*y3
  F2 =~ 0.7*y4 + 0.7*y5 + 0.7*y6
  F3 =~ 0.7*y7 + 0.7*y8 + 0.7*y9
  F1 ~~ 1*F1; F2 ~~ 1*F2; F3 ~~ 1*F3
  y1 ~~ 0.51*y1; y2 ~~ 0.51*y2; y3 ~~ 0.51*y3
  y4 ~~ 0.51*y4; y5 ~~ 0.51*y5; y6 ~~ 0.51*y6
  y7 ~~ 0.51*y7; y8 ~~ 0.51*y8; y9 ~~ 0.51*y9
'
model_fit <- 'F1 =~ y1+y2+y3; F2 =~ y4+y5+y6; F3 =~ y7+y8+y9'

p <- 9; k <- 3; q <- 24
df_model <- p*(p+1)/2 - q
p_star <- p*(p+1)/2; q_tilde <- p_star

sample_sizes <- c(50, 75, 100, 150, 200, 300, 500)
N_sim <- 2000
alpha <- 0.05

# Pre-compute critical value
crit_val <- qchisq(1 - alpha, df = df_model)
```

```r

# Results storage
results_mat <- matrix(NA, nrow = length(sample_sizes),
                      ncol = 6,
                      dimnames = list(
                        paste0("n=", sample_sizes),
                        c("Uncorrected", "Std1950", "Sav7", "Sav8", "Fixed", "N_valid")
                      ))

set.seed(42)

for (ni in seq_along(sample_sizes)) {
  n <- sample_sizes[ni]
  reject_count <- rep(0, 5)
  n_valid <- 0

  # Pre-compute multipliers for this n
  mult_std  <- 1 - (2*p + 4*k + 5) / (6 * (n - 1))
  numer_sav8 <- (2*p^2 + 3*p - 1) - q_tilde*(2*q_tilde^2 + 3*q_tilde - 1)
  mult_sav8 <- 1 - numer_sav8 / (12 * df_model * (n - 1))
  b_fix <- (2*p_star + 4*k - 2*q - 1) / (6 * (n - 1))
  c_fix <- 1 + b_fix / df_model

  for (sim in 1:N_sim) {
    dat <- tryCatch(
      simulateData(model_true, sample.nobs = n, empirical = FALSE),
      error = function(e) NULL
    )
    if (is.null(dat)) next

    fit <- tryCatch(
      cfa(model_fit, data = dat, estimator = "ML", warn = FALSE),
      error = function(e) NULL
    )
    if (is.null(fit)) next

    t_ml <- fitMeasures(fit, "chisq")
    if (is.na(t_ml)) next
    n_valid <- n_valid + 1

    # Uncorrected
    if (t_ml > crit_val) reject_count[1] <- reject_count[1] + 1
    # Std 1950
    if (t_ml * mult_std > crit_val) reject_count[2] <- reject_count[2] + 1
    # Savalei 7 (applied to ML as proxy)
    if (t_ml * mult_std > crit_val) reject_count[3] <- reject_count[3] + 1
    # Savalei 8
    if (t_ml * mult_sav8 > crit_val) reject_count[4] <- reject_count[4] + 1
    # Fixed
    if (t_ml / c_fix > crit_val) reject_count[5] <- reject_count[5] + 1
  }

  results_mat[ni, ] <- c(reject_count / n_valid, n_valid)
  cat(sprintf("n=%3d done: %d valid reps\n", n, n_valid))
}

# Display results
cat("\n")
cat(sprintf("%-8s  %12s  %8s  %8s  %8s  %8s  %7s\n",
            "n", "Uncorrected", "Std1950", "Sav7", "Sav8",
            "Fixed", "N_valid"))
cat(rep("-", 70), "\n")
for (ni in 1:nrow(results_mat)) {
  cat(sprintf("n=%-4d  %12.4f  %8.4f  %8.4f  %8.4f  %8.4f  %7d\n",
              sample_sizes[ni],
              results_mat[ni, 1], results_mat[ni, 2],
              results_mat[ni, 3], results_mat[ni, 4],
              results_mat[ni, 5], results_mat[ni, 6]))
}
cat(sprintf("\nNominal alpha = %.4f\n", alpha))
```

Listing 3: Multi-Sample-Size Simulation Script

4.4 Applying Corrections to a Real Dataset

The following script demonstrates how to apply all four corrections to a single SEM fit on a real dataset. It extracts the test statistic and model parameters, computes each corrected statistic, and reports the p-values.

```r
# ============================================================
# Apply All Bartlett Corrections to a Real Dataset
# Uses the HolzingerSwineford1939 dataset (built into lavaan)
# ============================================================

library(lavaan)

# Define a 3-factor CFA model for the HS dataset
# 9 observed variables: x1-x9
# 3 factors: visual, textual, speed
model <- '
  visual   =~ x1 + x2 + x3
  textual =~ x4 + x5 + x6
  speed    =~ x7 + x8 + x9
'

# Fit the model
fit <- cfa(model, data = HolzingerSwineford1939, estimator = "ML")

# Extract model information
summary_fit <- summary(fit, fit.measures = TRUE, standardized = TRUE)
t_ml <- fitMeasures(fit, "chisq")           # T_ML
df_model <- fitMeasures(fit, "df")           # degrees of freedom
p <- fitMeasures(fit, "nobs")               # will overwrite below
n <- fitMeasures(fit, "nobs")               # sample size
p_vars <- length(lavNames(fit))             # number of observed variables
k <- length(lavNames(fit, type = "lv"))     # number of latent variables

# Count free parameters
ptable <- parTable(fit)
q <- sum(ptable$free > 0)  # number of free parameters

# Compute p* and q_tilde
p_star <- p_vars * (p_vars + 1) / 2  # covariance structure
q_tilde <- p_star

cat("=====================================================\n")
cat("  BARTLETT CORRECTIONS FOR HolzingerSwineford1939\n")
cat("=====================================================\n\n")
cat(sprintf("Model:         %d-factor CFA\n", k))
cat(sprintf("Variables:     p  = %d\n", p_vars))
cat(sprintf("Factors:       k  = %d\n", k))
cat(sprintf("Sample size:   n  = %d\n", n))
cat(sprintf("Parameters:    q  = %d\n", q))
cat(sprintf("DF:            df = %d\n", df_model))
cat(sprintf("T_ML:             = %.3f\n\n", t_ml))

# Compute each corrected statistic
t_std  <- bartlett_std(t_ml, p_vars, k, n)
t_sav8 <- bartlett_sav8(t_ml, p_vars, n, df_model, q_tilde)
t_fix  <- bartlett_fixed(t_ml, p_vars, k, q, n, df_model)

# Compute p-values
p_uncorr <- 1 - pchisq(t_ml, df = df_model)
p_std    <- 1 - pchisq(t_std, df = df_model)
p_sav8   <- 1 - pchisq(t_sav8, df = df_model)
p_fix    <- 1 - pchisq(t_fix, df = df_model)

cat(sprintf("%-16s  %10s  %10s  %10s\n",
            "Method", "t*", "p-value", "Decision"))
cat(rep("-", 50), "\n")
cat(sprintf("%-16s  %10.3f  %10.6f  %10s\n",
            "Uncorrected", t_ml, p_uncorr,
            ifelse(p_uncorr < 0.05, "Reject", "Retain")))
cat(sprintf("%-16s  %10.3f  %10.6f  %10s\n",
            "Std 1950 (F1)", t_std, p_std,
```

```r
              ifelse(p_std < 0.05, "Reject", "Retain")))
cat(sprintf("%-16s  %10.3f  %10.6f  %10s\n",
            "Savalei 8 (F3)", t_sav8, p_sav8,
            ifelse(p_sav8 < 0.05, "Reject", "Retain")))
cat(sprintf("%-16s  %10.3f  %10.6f  %10s\n",
            "Fixed (F4)", t_fix, p_fix,
            ifelse(p_fix < 0.05, "Reject", "Retain")))

cat(sprintf("\nCritical value (alpha=0.05, df=%d): %.3f\n",
            df_model, qchisq(0.95, df = df_model)))
```

Listing 4: Applying Corrections to a Real SEM Fit

4.5 Computing Corrections Without lavaan

For users who compute their test statistics outside of lavaan (e.g., from Mplus, OpenMx, or custom estimators), the following standalone function computes all corrections given only the basic inputs.

```r
# ============================================================
# Standalone Bartlett Correction Calculator
# Works with any SEM software - just provide the inputs.
#
# Arguments:
#   t         : test statistic (chi-square value)
#   p         : number of observed variables
#   k         : number of latent factors
#   q         : number of free parameters
#   df        : degrees of freedom
#   n         : sample size
#   alpha     : nominal significance level (default 0.05)
#   stat_type : "ML" or "SB" (Satorra-Bentler)
#
# Returns: A list with all corrected statistics and p-values
# ============================================================

bartlett_all <- function(t, p, k, q, df, n, alpha = 0.05,
                         stat_type = "ML") {

  p_star <- p * (p + 1) / 2
  q_tilde <- p_star

  # F1: Standard 1950 (for ML)
  mult_std <- 1 - (2*p + 4*k + 5) / (6 * (n - 1))
  t_F1 <- t * mult_std

  # F2: Savalei 7 (same multiplier, for SB)
  t_F2 <- t * mult_std  # same formula, different input

  # F3: Savalei 8
  numer <- (2*p^2 + 3*p - 1) - q_tilde*(2*q_tilde^2 + 3*q_tilde - 1)
  mult_sav8 <- 1 - numer / (12 * df * (n - 1))
  t_F3 <- t * mult_sav8

  # F4: Fixed (for ML)
  b_fix <- (2*p_star + 4*k - 2*q - 1) / (6 * (n - 1))
  c_fix <- 1 + b_fix / df
  t_F4 <- t / c_fix

  # p-values
  crit <- qchisq(1 - alpha, df = df)

  out <- data.frame(
    Method = c("Uncorrected",
               "F1:Std1950",
               "F2:Savalei7",
               "F3:Savalei8",
               "F4:Fixed"),
    Statistic = c(t, t_F1, t_F2, t_F3, t_F4),
    P_value = c(1 - pchisq(t, df),
```

```
                  1 - pchisq(t_F1, df),
                  1 - pchisq(t_F2, df),
                  1 - pchisq(t_F3, df),
                  1 - pchisq(t_F4, df)),
    Reject = c(t, t_F1, t_F2, t_F3, t_F4) > crit,
    Correction = c("None",
                   sprintf("x%.4f", mult_std),
                   sprintf("x%.4f", mult_std),
                   sprintf("x%.4f", mult_sav8),
                   sprintf("/%.4f", c_fix)),
    stringsAsFactors = FALSE
  )

  attr(out, "crit") <- crit
  attr(out, "input") <- list(t=t, p=p, k=k, q=q, df=df, n=n)
  return(out)
}

# --- Example usage ---
# Suppose you fit a model in Mplus and got:
#   Chi-square = 85.3, df = 42, p = 12, k = 3, q = 33, n = 200
result <- bartlett_all(t = 85.3, p = 12, k = 3, q = 33,
                       df = 42, n = 200, alpha = 0.05)
print(result)
```

Listing 5: Standalone Correction Function (No SEM Package Required)

5 Understanding Type I Error Coverage

This section explains the theoretical and practical aspects of Type I error coverage that underlie the simulation framework.

5.1 What Is Type I Error and Why Does It Matter?

The Type I error rate of a hypothesis test is the probability of rejecting the null hypothesis H_0 when H_0 is true. For a test with nominal significance level α, the Type I error rate should be exactly α (or at most α). When the Type I error rate exceeds α, the test is *liberal*: it rejects too often, leading to false claims of model misfit. When it falls below α, the test is *conservative*: it has reduced power to detect genuine misfit.

In SEM, the standard practice is to compare the ML chi-square statistic T_{ML} to the χ^2_{df} critical value. This procedure has a Type I error rate of $\alpha + \mathcal{O}(n^{-1})$: the actual rejection probability differs from α by an amount that shrinks as $n \to \infty$, but can be substantial in small samples. The purpose of a Bartlett correction is to reduce this discrepancy to $\mathcal{O}(n^{-2})$, yielding accurate inference even at moderate sample sizes.

5.2 How the Simulation Works

The Monte Carlo simulation for Type I error coverage follows a four-step procedure:

Step 1: Generate data under H_0. We specify a population model that is exactly the model to be tested, generate random data from this model using the multivariate normal distribution with the model-implied covariance matrix $\boldsymbol{\Sigma}(\boldsymbol{\theta}_0)$, and record the sample size n. Because the data are generated from the model, the null hypothesis is true by construction.

Step 2: Fit the model. We fit the SEM to the generated data using maximum likelihood estimation, extract the test statistic $t = T_{\text{ML}}$, and compute the corrected statistics t^*_{Std}, t^*_{Sav8}, and t^*_{Fix} using the formulas from Section 2.

Step 3: Record rejection. For each statistic (uncorrected and corrected), we compare it to the critical value $\chi^2_{\text{df},1-\alpha}$ and record whether H_0 was rejected (statistic > critical value).

Step 4: Repeat and aggregate. We repeat Steps 1–3 for N_{sim} independent replications. The estimated Type I error rate for each method is the proportion of replications in which H_0

was rejected:

$$\hat{\alpha} = \frac{1}{N_{\text{sim}}} \sum_{i=1}^{N_{\text{sim}}} \mathbf{1}(t_i^* > \chi^2_{\text{df},1-\alpha}). \tag{34}$$

The standard error of this estimate is $\sqrt{\alpha(1-\alpha)/N_{\text{sim}}}$, which for $\alpha = 0.05$ and $N_{\text{sim}} = 2000$ is approximately 0.0049.

A well-calibrated test should have $\hat{\alpha} \approx 0.05$. Values substantially above 0.05 indicate a liberal test (too many false rejections), while values substantially below indicate a conservative test (insufficient power).

5.3 Interpreting Simulation Results

When interpreting the output of the simulation scripts, consider the following guidelines:

G1. Statistical uncertainty. With $N_{\text{sim}} = 2000$, the standard error is about 0.005. Differences smaller than 0.01 between methods may not be statistically meaningful. Report results to 4 decimal places but interpret differences of 0.02 or more as meaningful.

G2. Liberal vs. conservative. A test with $\hat{\alpha} = 0.08$ at $\alpha = 0.05$ rejects 60% more often than it should—a serious inflation. A test with $\hat{\alpha} = 0.03$ is conservative, missing 40% of the nominal rejections.

G3. Sample size trend. A good correction should show $\hat{\alpha} \to 0.05$ as n increases, and should maintain $\hat{\alpha} \approx 0.05$ even at small n. The uncorrected statistic typically shows $\hat{\alpha} > 0.05$ at small n, converging slowly to 0.05.

G4. Comparison across corrections. The Fixed correction (F4) should outperform the Standard (F1) at small n because it accounts for the full model structure. The Savalei corrections (F2, F3) applied to T_{SC} may show unpredictable behavior because they are ad hoc adaptations.

G5. Robustness checks. Vary the number of factors k, the factor loading magnitude, and the distribution of the data (normal vs. nonnormal) to assess the robustness of each correction. The Fixed correction is theoretically justified only under normality; the ad hoc corrections may work better under nonnormality purely by chance.

6 Numerical Examples with Expected Output

This section provides worked numerical examples showing the computation of each correction for specific parameter values, along with the expected Type I error patterns.

6.1 Example 1: Three-Factor CFA ($p = 9$, $k = 3$)

Consider a three-factor confirmatory factor analysis with $p = 9$ observed variables (3 per factor), factor loadings of 0.7, and $n = 150$. The model has $q = 24$ free parameters (9 loadings + 9 residual variances + 3 factor variances + 3 factor covariances), yielding $\text{df} = 45 - 24 = 21$.

Computation details:

$$\text{F1 multiplier} = 1 - \frac{2(9) + 4(3) + 5}{6(149)} = 1 - \frac{35}{894} = 0.9609, \tag{35}$$

$$\text{F3 numerator} = (2 \times 81 + 27 - 1) - 45(2 \times 2025 + 135 - 1) = 188 - 189045 = \text{very large}, \tag{36}$$

$$\text{F3 multiplier} = 1 - \frac{(2p^2 + 3p - 1) - \tilde{q}(2\tilde{q}^2 + 3\tilde{q} - 1)}{12 \times 21 \times 149}. \tag{37}$$

Table 2: Correction Factors for Example 1 ($p = 9$, $k = 3$, $n = 150$, df $= 21$)

Correction	**Factor c or Multiplier**	**Corrected t^* (for $t = 50$)**
F1: Standard 1950	$\times\, 0.9215$	46.075
F3: Savalei 8	$\times\, 0.8561$	42.803
F4: Fixed	$\div\, 1.0408$	48.041

For the Fixed correction:

$$b_{\text{fix}} = \frac{2(45) + 4(3) - 2(24) - 1}{6(149)} = \frac{90 + 12 - 48 - 1}{894} = \frac{53}{894} = 0.0593, \tag{38}$$

$$c_{\text{fix}} = 1 + \frac{0.0593}{21} = 1.0028. \tag{39}$$

6.2 Example 2: Single-Factor Model ($p = 6$, $k = 1$)

Consider a single-factor model with $p = 6$ indicators, $n = 100$, $q = 12$ (6 loadings + 6 residuals + 1 factor variance = 12, but with factor variance fixed to 1 for identification, $q = 12$). Then df $= 21 - 12 = 15$.

$$\text{F1 multiplier} = 1 - \frac{2(6) + 4(1) + 5}{6(99)} = 1 - \frac{21}{594} = 0.9646, \tag{40}$$

$$b_{\text{fix}} = \frac{2(21) + 4(1) - 2(12) - 1}{6(99)} = \frac{42 + 4 - 24 - 1}{594} = \frac{21}{594} = 0.0354, \tag{41}$$

$$c_{\text{fix}} = 1 + \frac{0.0354}{15} = 1.0024. \tag{42}$$

Note that for this simpler model, the Fixed correction factor is very close to 1, indicating that the ML statistic is already well-calibrated for this model structure.

7 Summary and Recommendations

Table 3: Decision Guide: Which Correction to Use

Scenario	**Recommended Correction**
Normal data, simple CFA	F1: Standard 1950 (well-established for factor models)
Normal data, general SEM	F4: Fixed (theoretically justified for general covariance structures)
Nonnormal data, using T_{SC}	F2 or F3 (ad hoc, but no better alternative currently exists)
Small sample ($n < 200$)	F4: Fixed (accounts for full model complexity)
Large sample ($n > 500$)	Any correction (impact is minimal at large n)
Uncertain about normality	Run simulation with both F1/F4 (on T_{ML}) and F2/F3 (on T_{SC})

The simulation framework provided in Section 4 allows practitioners to empirically evaluate which correction performs best for their specific model and data conditions. The key recommendation is to **always report both the uncorrected and corrected test statistics**, along with their p-values, so that readers can assess the sensitivity of the conclusions to the choice of correction.

References

References

[1] Bartlett, M. S. (1937). Properties of sufficiency and statistical tests. *Proceedings of the Royal Society A*, 160, 268–282.

[2] Bartlett, M. S. (1950). Tests of significance in factor analysis. *British Journal of Psychology, Statistical Section*, 3, 77–85.

[3] Cordeiro, G. M. and Cribari-Neto, F. (2014). *An Introduction to Bartlett Correction and Bias Reduction*. SpringerBriefs in Statistics, Springer.

[4] Hayakawa, T. (1977). The likelihood ratio criterion and the asymptotic expansion of its distribution. *Annals of the Institute of Statistical Mathematics*, 29, 359–378.

[5] Lawley, D. N. (1956). A general method for approximating to the distribution of likelihood ratio criteria. *Biometrika*, 43, 295–303.

[6] Savalei, V. (2010). Small sample statistics for incomplete nonnormal data: Extensions of complete data formulae. *Structural Equation Modeling*, 17(2), 241–264.

[7] Swain, A. J. (1975). A class of procedures for testing a covariance matrix under a standard condition. *Unpublished doctoral dissertation*, University of North Carolina at Chapel Hill.